A Pilot E_G Growth-Lensing Test of a Frozen Neutral-Lensing Growth Response in Reverse Universe Theory

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Abstract

Reverse Universe Theory (RUT) is treated here not as a complete replacement cosmology, but as motivation for a reduced and falsifiable growth-lensing bridge. The manuscript freezes a smooth transition function B(a), applies it to a nonrelativistic growth response mu(a), separates it from a lensing response Sigma(a), and compares the resulting fixed bridge cases with LambdaCDM using a pilot E_G growth-lensing compilation. Earlier background and growth pretests motivate, but do not independently establish, the carried-forward growth amplitude. The main 18-point diagonal E_G comparison favors the fixed neutral-lensing case, mu(a)=1+mu0 B(a), Sigma(a)=1, with Delta BIC=-2.108 relative to LambdaCDM, while full growth-lensing coupling is strongly disfavored. Because the E_G compilation is not covariance-clean and likely contains shared survey inputs, the primary robustness result is a leave-one-family-out stress test rather than a point-jackknife: the neutral-lensing result remains negative in Delta BIC for all eight family-drop tests, with values ranging from -0.887 to -2.249. A matched constant-mu control narrows the interpretation. A fixed constant growth offset slightly outperforms the fixed RUT bridge by Delta BIC=0.157, so the current pilot data support a neutral-lensing growth response but do not uniquely resolve the RUT tanh activation profile. A first-pass comparison with published mu/Sigma constraints indicates that the redshift-activated growth amplitude is not obviously excluded, but a full likelihood-level mapping remains required. The result is presented as a reproducible pilot signal and a target for covariance-controlled, frozen-model follow-up testing, not as detection-level evidence or a replacement of LambdaCDM.

Keywords: cosmology; modified gravity; growth of structure; gravitational lensing; E_G statistic; LambdaCDM; model comparison; Bayesian Information Criterion; Reverse Universe Theory

One-sentence claim: A frozen neutral-lensing growth response improves a pilot diagonal E_G comparison against LambdaCDM and survives family-drop robustness checks, but the present data do not yet distinguish the RUT bridge activation profile from a simple fixed growth offset; covariance-clean testing remains required.

1. Submission posture and claims discipline

This manuscript is a pilot growth-lensing submission draft. It is not submitted as a final cosmological replacement claim. Its contribution is the transparent testing of a reduced bridge model, including negative results, matched controls, and explicit limitations. The primary claim is deliberately narrow: the pilot E_G data favor a neutral-lensing growth response over LambdaCDM under the reduced estimator, while full growth-lensing coupling is strongly disfavored.

Allowed claim	Claim to avoid
RUT is testable through a reduced bridge model.	RUT is proven as a final theory of cosmology.

Allowed claim	Claim to avoid
Fixed neutral-lensing RUT improves on LambdaCDM in pilot diagonal E_G tests.	RUT has definitively replaced LambdaCDM.
The E_G result survives family-reduction, family-weighting, point-drop, and family-drop stress tests.	The E_G result is covariance-clean or confirmed.
Matched controls show the bridge is competitive but not uniquely distinguished.	The tanh bridge shape is already proven by current data.
Full growth-lensing coupling is disfavored in the pilot E_G lane.	Any RUT-shaped modification is favored.

2. Phenomenological motivation

The present manuscript does not require the reader to accept the full RUT framework. It tests only a reduced observational structure: a bridge function B(a), a growth response mu(a), a lensing response Sigma(a), and the E_G statistic as a growth-lensing diagnostic. The purpose of the broader RUT motivation is limited to one modeling choice: growth and lensing are allowed to be related through the same bridge without being forced to respond identically.

In observational cosmology, light propagation, redshift, angular diameter distance, luminosity distance, time delay, lensing shear, and clustering statistics already serve as operational probes of geometry. This paper asks whether a smooth bridge in scale factor improves a growth-lensing consistency comparison relative to LambdaCDM. The conceptual language is not treated as evidence; from the model section onward the test is carried by explicit functions, fixed parameter conventions, chi-square, AIC/BIC, and robustness checks.

Submission-safe framing: The phrase "RUT bridge" means a phenomenological, testable transition parameterization. It does not imply that the full ontology of RUT has been empirically established.

3. Minimal bridge model and tested cases

3.1 Reduced bridge variable

The reduced bridge is written as a smooth transition in scale factor:

B(a) = 1/2 [1 + tanh((a - a_t)/Delta a)]

where a is the scale factor, a_t is the transition scale, and Delta a controls the transition width. In the Sprint A-C tests summarized here, the reduced bridge uses a_t = 0.55 and Delta a = 0.08. This is not presented as a final theory parameterization; it is a controlled pilot form that allows comparison with LambdaCDM under common model-selection statistics.

3.2 Background response

At the background level, the bridge modifies the effective dark-energy equation of state through a small radiation-like deformation:

w_DE(a) = -1 + beta B(a)/3

The factor 1/3 is used as a phenomenological equation-of-state limit in the reduced test. It is not presented as literal photon domination. Sprint A background tests found the bridge stable and mockdetectable, but real cosmic-chronometer and BAO background data did not prefer it over LambdaCDM after parameter penalties.

3.3 Growth and lensing responses

For structure growth, the bridge is introduced through:

mu(a) = 1 + mu0 B(a).

Here mu(a) represents the effective nonrelativistic growth response relative to the general-relativistic baseline. LambdaCDM corresponds to mu(a)=1. The Sprint B joint background-plus-growth pretest found a small positive growth response near mu0 = 0.173. This value is carried forward as a fixed input for the E_G analysis rather than fitted at the E_G stage.

The lensing channel is separated from the growth channel through:

Sigma(a) = 1 + lambda_Sigma mu0 B(a).

Case	Definition	Role in paper
LambdaCDM	mu(a)=1; Sigma(a)=1	Baseline comparison model.
RUT full coupling	mu(a)=1+mu0 B(a); Sigma(a)=1+mu0 B(a)	Fixed RUT case; strongly disfavored in pilot E_G tests.
RUT neutral lensing	mu(a)=1+mu0 B(a); Sigma(a)=1	Main fixed RUT case; no extra E_G-stage parameter.
RUT lambda-fit	mu(a)=1+mu0 B(a); Sigma(a)=1+lambda_Sigma mu0 B(a)	Diagnostic case; explores preferred lensing-channel response.
Constant-mu fixed control	mu(a)=1+mu_c; Sigma(a)=1; mu_c=0.172759	Matched fixed control against generic growth rescaling.
Constant-mu fitted control	mu(a)=1+mu_c; Sigma(a)=1; mu_c fitted	Diagnostic control; pays one E_G-stage parameter.

3.4 Freeze statement

For the frozen comparison, the reduced bridge model is fixed as:

mu(a) = 1 + mu0 B(a),   Sigma(a) = 1,   a_t=0.55,   Delta a=0.08.

The value mu0 is carried forward from the Sprint B growth pretest rather than refit to the E_G data. Parameter scans, alternate bridge centers, or fitted lambda_Sigma values are reported only as secondary diagnostics. This freeze statement is intended to prevent the E_G stage from becoming a post-hoc optimization exercise.

4. Background and growth pretests

The background and growth pretests motivate the E_G analysis without claiming that the background or growth-only sectors establish RUT. They define the reduced bridge, show where it does and does not improve on LambdaCDM, and justify why a growth-lensing consistency observable becomes the central target.

Pretest	Dataset / observable	Key result	Interpretation
Sprint A	Cosmic chronometers, BAO-style distance ratios, broad	beta approx 0.1006; Delta chi2 approx -0.078; Delta	Stable and mock-validated, but background data prefer

Pretest	Dataset / observable	Key result	Interpretation
	r_d prior	BIC approx +3.71	LambdaCDM.
Sprint B growth-only	Pilot f sigma_8 compilation	Small positive growth modification allowed; not BIC-supported	Growth response is stable but weak.
Sprint B joint	Background + f sigma_8	mu0 approx 0.173; Delta chi2 approx -0.476; Delta BIC approx +3.63	Mild growth-sector improvement, but not enough to earn BIC support.

The pretests imply that the reduced bridge is not primarily a background-expansion correction. The carried-forward mu0 value is methodologically useful because it freezes the E_G test before evaluation, but it is also statistically fragile because the growth-only sector did not earn BIC support. This motivates the mu0 sensitivity and constant-mu control checks below.

5. The E_G observable and pilot dataset

5.1 Purpose of the E_G test

The E_G statistic combines information from galaxy clustering, redshift-space growth, and lensing. In the reduced bridge language, it tests whether the growth channel and lensing channel respond identically, separately, or oppositely to the bridge. The E_G lane is therefore used as a growth-lensing consistency diagnostic.

5.2 Working E_G approximation and estimator caveat

For the present pilot comparison, the model prediction is constructed through the phenomenological approximation:

E_G(z) approx Omega_m0 Sigma(a) / f(z),   a = 1/(1+z).

This is a reduced estimator-level comparison, not a reanalysis of raw survey pipelines. Published E_G-style measurements differ in angular scale cuts, fiducial cosmology, galaxy-bias treatment, redshift-space distortion inputs, lensing maps, and reconstruction assumptions. This heterogeneity is not a cosmetic issue: family-dependent estimator offsets could be partially degenerate with the inferred lambda_Sigma response and could mimic or suppress a neutral-lensing preference. The one-per-family, family-weighted, and family-drop tests are therefore partial stress tests of estimator heterogeneity, not substitutes for re-deriving all E_G values under a common likelihood and covariance model.

Estimator limitation: The E_G result should be read as a reduced observational consistency comparison using published E_G values. It is not a covariance-clean or pipeline-reprocessed measurement.

5.3 Pilot E_G compilation

The pilot compilation contains 18 E_G-style measurements grouped into eight measurement families. The compilation is intentionally broad: it tests whether the neutral-lensing signal appears only in one family or persists across a mixed growth-lensing dataset.

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Family	Points	Role in pilot compilation
Reyes	1	Classic low-redshift anchor near z approx 0.32.
Blake	2	Historical E_G bins.
Amon	3	Multiple lower-redshift bins.
Alam	1	Single z approx 0.57 comparison point.
Pullen	1	CMB-lensing cross-correlation style point near z approx 0.57.
Wenzl	2	Recent harmonic-space precision E_G measurements.
Grimm	4	Recent Weyl-potential/velocity redshift-bin set.
Li/Xia	4	Recent binned lensing/growth comparison set; published as ApJS 276, 71.

5.4 Diagonal-error status and effective information content

The pilot compilation uses diagonal errors. This is the main statistical limitation and should be treated seriously. Several E_G-style measurements may share Planck lensing maps, BOSS or SDSS galaxy samples, RSD inputs, fiducial assumptions, or calibration systematics. Therefore the effective number of independent constraints may be closer to the number of families, or even fewer correlated blocks, than to the 18 individual points. Point-level jackknife tests are useful for detecting single-point leverage, but they do not remove shared covariance. The leave-one-family-out check is therefore treated as the primary robustness stress test, with the point-jackknife presented as secondary stability evidence.

In the extreme limit where all measurement families were treated as one fully correlated block, the effective information content would approach n_eff = 1 and an ordinary chi-square model comparison would no longer be interpretable. The family-drop and family-compression results are therefore practical lower-bound robustness checks, not substitutes for a covariance-clean likelihood.

Model	E_G-stage k	BIC interpretation
		penalty.
Constant-mu fixed	0	Matched fixed control.
Constant-mu fitted	1	Diagnostic; pays one fitted-parameter penalty.

Effective-n caveat: If the true independent information content is closer to the number of correlated families than to 18 points, the fitted-model BIC penalties are approximate. This does not affect fixed-vs-fixed comparisons because k is equal, but it matters for diagnostic fitted models.

7. Pilot E_G results

The central fixed RUT test is the neutral-lensing case, lambda_Sigma=0, because it introduces no additional E_G-stage fitted parameter. The working model uses the Sprint B growth amplitude mu0 approx 0.173 and evaluates LambdaCDM, full coupling, neutral lensing, and lambda-fit cases through the same pipeline.

Model	lambda_Sigma	Sigma0	chi2	k_EG	Delta BIC vs LambdaCDM	Delta AIC vs LambdaCDM
LambdaCDM	-	-	20.289174	0	0.000000	0.000000
RUT full coupling	1.000000	0.172759	56.318530	0	+36.029356	+36.029356
RUT neutral	0.000000	0.000000	18.181581	0	-2.107592	-2.107592
RUT anchor lambda	-0.426396	-0.073664	18.488367	0	-1.800807	-1.800807
RUT lambda-fit	-0.199992	-0.034550	17.092026	1	-0.306776	-1.197148

On the full 18-point diagonal pilot compilation, fixed neutral-lensing RUT improves the raw chi-square relative to LambdaCDM by Delta chi2 = -2.107592. Because the neutral-lensing case is treated as a fixed prediction in the E_G stage, it carries no additional E_G-stage parameter penalty, so Delta BIC = -2.107592. This is a weak-to-moderate pilot preference, not a decisive result.

Full growth-lensing coupling fails strongly. The lambda_Sigma=1 case gives Delta BIC = +36.029356 relative to LambdaCDM. This failure is scientifically useful: the E_G data do not reward arbitrary RUT-shaped modifications. The data prefer separated growth-lensing response rather than identical growth and lensing activation.

The lambda_Sigma-fitted diagnostic lowers raw chi-square but pays an information-criterion penalty. Its best-fit lambda_Sigma is near -0.200, suggesting neutral to mildly softened lensing relative to growth, but the BIC advantage shrinks to Delta BIC approx -0.307. It is therefore diagnostic rather than headline evidence.

8. Robustness and matched-control checks

8.1 Family-drop check as primary robustness result

Because shared covariance is the main weakness of the pilot compilation, the leave-one-family-out check is the most credible robustness test currently included. It removes an entire measurement family at a time and reruns the fixed neutral-lensing comparison against LambdaCDM. This directly

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The Grimm family is particularly informative because its Weyl-potential/velocity construction is among the closest entries in the compilation to directly testing the lensing-response side of the mu-Sigma split; dropping Grimm still yields Delta BIC = -2.067.

Dropped family	n	LCDM chi2	RUT neutral chi2	Delta BIC	Result
Alam	17	19.929872	17.681073	-2.248798	RUT wins
Amon	15	12.214124	11.327549	-0.886575	RUT wins
Blake	16	18.596344	16.537899	-2.058445	RUT wins
Grimm	14	18.796840	16.730284	-2.066556	RUT wins
Li/Xia	14	18.533232	16.371549	-2.161682	RUT wins
Pullen	17	14.332923	12.707392	-1.625531	RUT wins
Reyes	17	19.766657	17.954331	-1.812326	RUT wins
Wenzl	16	19.854225	17.960992	-1.893233	RUT wins

The fixed neutral-lensing bridge remains preferred in 8/8 leave-one-family-out tests. The weakest surviving case occurs when Amon is removed, with Delta BIC = -0.886575. The strongest case occurs when Alam is removed, with Delta BIC = -2.248798. This does not make the result covariance-clean, but it shows that no single measurement family is necessary for the pilot advantage to remain negative in Delta BIC.

8.2 Family compression and point-jackknife checks

The full diagonal compilation gives Delta BIC = -2.108. The one-per-family minimum-error subset gives Delta BIC = -1.926, and the family-weighted average subset gives Delta BIC = -1.963. The leave-one-point-out jackknife gives an 18/18 RUT win fraction with worst Delta BIC = -1.266. The point-jackknife is reassuring but limited: it tests sensitivity to individual points, not to correlated blocks. Therefore it is treated as secondary to the family-drop check.

The fixed constant-mu control very slightly outperforms the fixed RUT bridge by Delta BIC(RUT - constant mu fixed) = +0.157315, which is negligible on the scale of model-selection evidence. This near-tie is scientifically important: at current pilot precision, the data indicate that a neutral-lensing growth response is favored over LambdaCDM, but they do not determine whether the response is better described by the RUT redshift-activated bridge or by a redshift-independent growth offset. The fitted constant-mu diagnostic obtains lower raw chi-square, but after the one-parameter BIC penalty gives Delta BIC = -0.706 against LambdaCDM, weaker than the fixed RUT bridge at Delta BIC = -2.108. Thus the bridge remains competitive, but bridge uniqueness is not established.

9. Interpretation: neutral lensing, bridge uniqueness, and what is actually learned

The tested pattern is specific. RUT does not gain its strongest support from background expansion tests, and full growth-lensing coupling is disfavored by the pilot E_G comparison. The strongest fixed RUT case occurs when the growth channel carries a small bridge response while the lensing channel remains neutral.

In many modified-gravity interpretations, a stronger gravitational response would be expected to affect both matter motion and light bending. The pilot result suggests a more constrained possibility: matter clustering can be slightly sensitive to the bridge while light-deflection geometry remains close to the LambdaCDM value. Neutral lensing is therefore an empirically motivated boundary case rather than an arbitrary simplification.

The matched constant-mu control clarifies the present evidence boundary. At current pilot precision, the data favor a neutral-lensing growth response, but they do not decisively distinguish the RUT activation profile from a simpler constant growth rescaling. The bridge-specific prediction is therefore not merely that growth is enhanced. It is that the enhancement follows the redshift-dependent activation profile B(a), with a transition width and timing that should become testable when covariance-clean E_G measurements are available across multiple redshift bins. A constant-mu model has no corresponding timing structure, so future tests should focus on whether the residual pattern tracks the predicted activation shape rather than only the average amplitude.

Current interpretation: The pilot E_G evidence points toward growth-lensing channel separation. It does not yet prove that the tanh bridge shape is uniquely preferred. A covariance-clean multi-redshift analysis should treat bridge shape as the next discriminant.

10. External constraints and mu0 sensitivity

10.1 First-pass mu-Sigma compatibility check

The number mu0 approx 0.173 should not be read as a constant 17.3 percent enhancement at all redshifts. In the frozen bridge, the active deviation is mu(z)-1 = mu0 B(a), where B(a)=1/2[1+tanh((a-a_t)/Delta a)], a_t=0.55, Delta a=0.08, and a=1/(1+z). This redshift activation matters when comparing the pilot bridge to published modified-gravity constraints.

z	a	B(a)	mu(z)-1	mu(z)
0.270	0.787	0.997	0.1725	1.1725
0.320	0.758	0.994	0.1717	1.1717

z	a	B(a)	mu(z)-1	mu(z)
0.470	0.680	0.963	0.1663	1.1663
0.570	0.637	0.898	0.1551	1.1551
0.630	0.613	0.830	0.1434	1.1434
0.771	0.565	0.591	0.1022	1.1022
0.800	0.556	0.535	0.0925	1.0925

A direct one-to-one comparison is not exact because published modified-gravity analyses define mu and Sigma with different redshift/scale dependences, priors, datasets, and background assumptions. Still, representative mu-Sigma constraints provide a useful first-pass compatibility check. Andrade et al. report broad growth-side constraints such as mu0-1 = 0.02 +/- 0.19 and 0.07 +/- 0.18 depending on CMB combinations. Ishak et al. report a DESI full-shape modified-gravity constraint around mu0 = 0.05 +/- 0.22 for one representative dataset combination. Under these broad comparisons, a rough amplitude near 0.173 is not obviously excluded, although this does not replace a full likelihood-level mapping.

The headline neutral-lensing case has Sigma(a)-1=0. This is important because Sigma-side constraints are generally tighter than growth-side constraints, and the pilot E_G data strongly reject full positive growth-lensing coupling. The compatibility statement is therefore limited: the frozen growth amplitude is not obviously excluded as a rough growth-side comparison, while the neutral-lensing assumption avoids the stronger lensing-side tension that would arise if Sigma were forced to track mu.

Constraint boundary: This is a first-pass compatibility check. A JCAP-level version should still map the RUT bridge into the specific time- and scale-dependent mu-Sigma parameterization used by each external likelihood.

10.2 Linearized mu0 sensitivity diagnostic

Because the carried-forward mu0 value comes from a growth pretest that did not earn BIC support over LambdaCDM, the pilot E_G result should not depend too sharply on the exact value mu0 = 0.172759. As a first-pass diagnostic, the official C10 prediction table was linearly interpolated between the LambdaCDM prediction and the fixed RUT neutral prediction to estimate Delta BIC for several mu0 values. This is not a substitute for rerunning the full growth solver at each value, but it tests whether the result appears fine-tuned to the carried-forward amplitude.

mu0 value	Approx chi2	Approx Delta BIC vs LambdaCDM	Reading
0.000	20.289174	0.000000	LambdaCDM limit.
0.050	19.510760	-0.778413	Weak improvement.
0.100	18.869553	-1.419621	Negative Delta BIC persists.
0.150	18.365552	-1.923622	Close to headline result.
0.172759	18.181581	-2.107592	Frozen headline value.
0.200	17.998757	-2.290417	Slightly stronger under linearized diagnostic.
0.250	17.769168	-2.520005	Still negative.
0.300	17.676786	-2.612388	Still negative; suggests amplitude is not sharply

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This linearized diagnostic suggests that the negative Delta BIC is not sharply localized at mu0 = 0.172759 and does not disappear under modest changes in the carried-forward amplitude. However, this table is a linearized sensitivity diagnostic, not a replacement for a full re-solve of the growth equation and covariance-aware likelihood at each mu0 value. Although the linearized scan becomes more favorable at larger mu0, this should not be interpreted as a measured preference for larger amplitudes because f(z) depends nonlinearly on the integrated growth history; a submission-grade scan should rerun the full growth solver at each mu0.

11. Limitations and publication-grade requirements

11.1 Current evidential status

The central result is meaningful because neutral lensing uses no additional E_G-stage fitted parameter. However, the evidence level remains weak-to-moderate. The correct interpretation is not that RUT has displaced LambdaCDM, but that one specific growth-lensing channel has produced a repeatable pilot advantage that justifies a more rigorous test.

Potential outcome	Interpretation
Covariance-clean or strict-independence subset gives Delta BIC > 0.	The current pilot signal is subset-sensitive or not robust.
Matched constant-mu or other controls reproduce the result with equal or fewer assumptions.	The signal is a generic growth-response effect, not bridge-specific.
Independent growth-lensing observables require Sigma(a) to track mu(a).	The neutral-lensing bridge is weakened.
Frozen model remains negative in Delta BIC under covariance-clean handling.	The pilot result moves toward stronger evidence.
Bridge-shape tests distinguish B(a) from constant-mu across redshift bins.	The RUT-specific bridge claim strengthens.

12. Conclusion and next work

This paper has not attempted to establish the full Reverse Universe Theory. Its narrower purpose was to test whether a reduced RUT bridge leaves a measurable signature in the relation between growth and lensing using the E_G statistic as a pilot observable. The resulting conclusion is cautious: RUT does not yet have a confirmed observational victory over LambdaCDM, but the neutral-lensing bridge has produced a hardened pilot advantage in the E_G lane.

The central empirical result is that a fixed neutral-lensing growth response improves the pilot E_G comparison relative to LambdaCDM without introducing an extra E_G-stage fitted parameter. In the full diagonal compilation, Delta BIC(RUT neutral - LambdaCDM) = -2.107592. The advantage remains in one-per-family and family-weighted reductions, and the strongest robustness statement is the family-drop result: fixed neutral-lensing RUT remains preferred in all 8/8 leave-one-family-out tests, with Delta BIC ranging from approximately -0.887 to -2.249.

The matched-control result narrows the claim. A fixed constant-mu growth offset slightly outperforms the fixed RUT bridge by Delta BIC = 0.157, a negligible difference that prevents a unique bridge-shape claim at current precision. Conversely, the fixed RUT bridge outperforms a fitted constant-mu diagnostic after BIC penalty. The correct conclusion is therefore that the pilot data support a neutral-lensing growth response and motivate testing the RUT redshift-activation shape, not that the tanh bridge has already been uniquely selected.

The best current reduced data-facing form remains:

mu(a) = 1 + mu0 B(a),   Sigma(a) = 1.

The next decisive step is a covariance-clean E_G analysis with the model frozen before evaluation, matched controls included from the start, survey-family overlap explicitly audited, and bridge-shape sensitivity tested against constant-mu and other smooth controls. If the fixed neutral-lensing result remains favored under that protocol, RUT would move from hardened pilot evidence toward a stronger growth-lensing evidence claim. If it fails, that result would still be useful as a constraint on the bridge.

Final claim: The fixed neutral-lensing bridge deserves covariance-clean growth-lensing testing; this paper does not claim that RUT replaces LambdaCDM.

Appendix A. Run 10 covariance-ready diagnostic

Run 10 is included as a secondary diagnostic, not as a replacement for the 18-point fixed neutral-lensing result. It uses a four-point real E_G subset and a different bridge-search setup. It separates three ideas: a positive residual-shape lead near z approximately 0.47, a free-fit candidate near z approximately 1.5, and true model preference after scan penalties. Because its bridge parameters differ from the main frozen model, it is not part of the headline comparison.

Model	Key result	Interpretation
LambdaCDM/null proxy	chi2=2.7317; Delta BIC=0	Baseline remains safest reference.
Locked bridge near z approx 0.47	A_EG=-0.0143; chi2=2.7224; Delta BIC=+1.377	Shape survives, but model preference is not strong enough.
Best free bridge, local penalty	a*=0.40, Delta a=0.04, z* approx 1.5; Delta BIC=-0.843	Better fit under local scoring only.
Best free bridge, global scan penalty	same candidate; Delta BIC=+1.930	Does not improve relative to LambdaCDM after scan freedom.

The strict interpretation is that the free bridge can find a better-fitting candidate under local scoring, but the advantage is not robust to the global scan penalty. The locked z approx 0.47 template shows strong residual-shape correlation with shuffle p approximately 0.0399 and positive leave-one-out correlations. Because the subset is small, this is directional stability rather than evidence-grade significance.

Appendix B. Bridge-state transition diagnostic

This appendix preserves the bridge-state diagnostic as supplementary material rather than a main-paper claim. The bridge activation is B(a)=1/2[1+tanh((a-a*)/Delta a)] with z* = 0.35, a* approximately 0.7407, and Delta a = 0.03. The effective response model produced state-transition behavior: residual amplitude tracks B(a), while the derivative of the residual tracks dB/da.

The diagnostic found correlations between the absolute derivative of Delta E_G and dB/da around 0.91-0.97, with the derivative peak near z approximately 0.3409, consistent with the bridge midpoint. Null-control tests compared the true bridge template against wrong centers, wrong widths, and shuffled templates; the true template ranked first across tested k-values. A 250-trial parameter-prior robustness scan found the strongest window near k = 0.20-0.30 h/Mpc, with pass rates of 74.4% at k=0.20 and 80.4% at k=0.30.

This diagnostic is not discovery-level evidence. Its submission-safe role is to motivate future covariance-aware E_G, RSD, weak-lensing, and BAO likelihood comparisons. It should not compete with the main fixed neutral-lensing result.

Appendix C. Data and code availability statement

The accompanying replication packet should include the pilot E_G CSV, background and growth pretest result tables, hardened subset comparison table, leave-one-point-out jackknife table, leave-one-family-out table, matched constant-mu control tables, Run 10 diagnostic files, and a verification script or notebook that reproduces the headline chi-square, AIC, and BIC values. The expected

Appendix C. Data and code availability statement

The accompanying replication packet should include the pilot E_G CSV, background and growth pretest result tables, hardened subset comparison table, leave-one-point-out jackknife table, leave-one-family-out table, matched constant-mu control tables, Run 10 diagnostic files, and a verification script or notebook that reproduces the headline chi-square, AIC, and BIC values. The expected verification output should be printed in the README so readers can compare their local run against the manuscript tables. The packet should also include the constant-mu control outputs, bridge-shape family-drop diagnostic, and the mu0 sensitivity table introduced in this version.

For journal submission, the public archive link, version tag, and DOI should be inserted here only after the replication packet is archived. The manuscript should not rely on private conversation logs or undocumented scripts as evidence.

Appendix D. Covariance-clean follow-up protocol and survey audit

The model should be frozen before any covariance-clean or strict-independence follow-up. The frozen values are a_t=0.55, Delta a=0.08, mu0 approximately 0.173, lambda_Sigma=0. No parameter in the first follow-up gate should be refit. The purpose is to test whether the neutral-lensing signal survives a cleaner likelihood, not to discover a new optimum after seeing cleaner data.

Follow-up gate	Action	Pass/fail meaning
Gate 1: family audit	Identify overlap, shared maps, and shared RSD/lensing inputs.	If overlap cannot be documented, the result stays pilot-level.
Gate 2: primary subset	Use covariance blocks where available or one strict non-overlapping point per family.	If Delta BIC remains negative, the pilot survives first independence hardening.
Gate 3: frozen holdout	Do not refit a_t, Delta a, mu0, or lambda_Sigma.	A positive result becomes more meaningful because it is not tuned to the E_G data.
Gate 4: matched controls	Compare against constant-mu, CPL-like transition, and simple modified-gravity controls.	This distinguishes a RUT-specific bridge from generic flexibility.

Appendix E. Model-discrimination roadmap

The present pilot result motivates a focused follow-up program aimed at separating three possibilities: a generic growth-rescaling effect, a redshift-dependent bridge response, and a covariance artifact caused by overlapping E_G measurements. The following roadmap states the minimum tests required before the bridge shape can be claimed as distinctive.

1. Reproducible archive. The follow-up analysis should provide the E_G data table, official prediction table, hardened comparison tables, leave-one-family-out table, constant-mu control outputs, mu0 sensitivity diagnostic, and a one-command verification script in a public archive.

2. Covariance and overlap control. The primary likelihood should use published covariance blocks where available. Where full covariance is unavailable, the analysis should report family-level compression and block-drop tests as conservative alternatives.

3. Bridge-shape discrimination. The frozen RUT bridge should be compared against fixed and fitted constant-mu controls, shifted bridge centers, shuffled bridge timing, and at least one smooth non-RUT transition model. The key question is whether residuals track B(a), not merely whether average growth is enhanced.

4. Amplitude sensitivity. The mu0 sensitivity scan should be repeated with a full growth re-solve at each amplitude, not only a linearized diagnostic. If the negative Delta BIC survives across a reasonable range of amplitudes, the result is less likely to be a fine-tuned artifact of the Sprint B carry-forward value.

5. Independent observables. If the effect is real, it should leave a consistent imprint in related growth-lensing observables such as RSD-informed growth reconstructions, weak-lensing shear combinations, CMB-lensing cross-correlations, and future E_G measurements at multiple redshifts.

6. Falsification rule. If covariance-clean data favor LambdaCDM, if the sign of Delta BIC reverses under conservative family compression, or if matched controls reproduce the signal without the bridge timing, the present result should be interpreted as a generic growth-response hint or a covariance-limited artifact rather than bridge-specific evidence.

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Appendix F. Supplementary bridge-shape family-drop diagnostic

A supplementary bridge-shape check was added after the v12 family-drop review. The test repeats the residual-gap comparison after removing one E_G measurement family at a time. The plotted quantity is Delta BIC(tanh minus constant). Negative values favor the frozen RUT tanh-gap shape; positive values favor the constant-gap control.

Figure F1. Leave-one-family-out bridge-shape stability (negative values indicate that the frozen tanh bridge outperforms the constant-gap control; positive values indicate that the constant-gap control outperforms the frozen tanh bridge). The tanh shape beats the constant-gap control only when Li/Xia is removed. In the other seven drops, the constant-gap control slightly wins. However, the differences are very small: Delta BIC ranges from -0.117 to +0.293, so the diagnostic supports a near-tie rather than a decisive shape selection.

Dropped family	**Delta BIC tanh - constant**	**Preferred shape**
Li/Xia	-0.117	tanh
Blake	+0.083	constant
Pullen	+0.110	constant
Grimm	+0.118	constant
Wenzl	+0.153	constant
Alam	+0.160	constant
Reyes	+0.180	constant
Amon	+0.293	constant

Interpretation. This diagnostic strengthens the conservative v12 conclusion. The pilot E_G residuals favor a nonzero neutral-lensing growth gap over the null comparison, but current diagonal-error data do not resolve whether the gap is genuinely tanh-shaped or approximately constant. The RUT-specific prediction is therefore sharpened rather than claimed as already proven: future covariance-clean, multi-redshift E_G data should determine whether the residual gap tracks the frozen activation profile B(a) instead of remaining a redshift-independent growth offset.

Submission use. Figure F1 should be treated as a supplementary diagnostic supporting the statement that bridge-shape uniqueness is the next discriminant, not as headline evidence for the tanh bridge.

Appendix G. Final referee-proofing notes for the frozen pilot analysis

This appendix records final pre-submission clarifications added after the v12 bridge-shape diagnostic. These notes do not introduce a new fit; they clarify the statistical boundary conditions, estimator interpretation, and source status for the frozen neutral-lensing pilot analysis.

- **Effective-n limit.** In the extreme limit where all E_G measurement families were treated as one fully correlated block, the effective information content would approach n_eff = 1 and a chi-square model comparison would no longer be interpretable. The family-drop and family-compression tests are therefore practical robustness lower bounds, not substitutes for a covariance-clean likelihood.
- **Grimm estimator check.** The Grimm family is especially informative because its Weyl-potential/velocity construction is among the closest entries in the compilation to directly testing the lensing-response side of the mu-Sigma split. Dropping Grimm still leaves the neutral-lensing bridge favored over LambdaCDM with Delta BIC = -2.067, strengthening the estimator-robustness interpretation.
- **mu0 sensitivity caveat.** Because f(z) depends nonlinearly on the integrated growth history, the linearized mu0 interpolation in Section 10.2 is approximate. A submission-grade follow-up should rerun the full growth solver over the stated mu0 grid rather than relying only on linear interpolation.
- **Appendix B parameter mismatch.** Appendix B used an earlier exploratory residual-shape bridge with z* = 0.35, a* approximately 0.7407, and Delta a = 0.03 before the main frozen bridge parameterization was fixed at a_t = 0.55 and Delta a = 0.08. It is preserved only as an exploratory diagnostic and should not be read as a competing headline model.
- **Li/Xia source status.** The Li/Xia measurements used in the pilot compilation are no longer treated as uncertain-status points: the paper is available as arXiv:2501.02852 and is listed as published in Astrophysical Journal Supplement Series 276, 71. The remaining caution concerns covariance/overlap handling, not publication status.
- **Replication packet status.** The self-referential replication-packet placeholder has been removed from the formal reference list. The public archive DOI/version should be inserted in the Data and Code Availability statement only after the packet is archived, or supplied separately as supplementary material during submission.

Submission note. The formal manuscript should keep the main claim exactly as stated in v12: the fixed neutral-lensing growth response is a reproducible diagonal-error pilot result that motivates covariance-clean testing. The tanh bridge shape remains a prediction and next discriminant, not an established detection.

Appendix H. Full nonlinear mu0 growth-solver sensitivity diagnostic

As a follow-up to the initial linearized interpolation check in Section 10.2, the neutral-lensing growth equation was re-solved at each value in the grid mu0 = {0.00, 0.05, 0.10, 0.15, 0.172759, 0.20, 0.25, 0.30}. The bridge shape was kept frozen at a_t = 0.55 and Delta a = 0.08, with Sigma(a) = 1 throughout. This test checks whether the mu0 sensitivity is merely an artifact of linear interpolation.

Baseline-comparison note: the reconstructed verification pipeline gives LambdaCDM chi2 = 27.044782, while the official C10 headline table gives LambdaCDM chi2 = 20.289174. These absolute baselines differ because Appendix H recomputes E_G directly through the local full growth solver and estimator implementation rather than reading from the frozen C10 prediction table. Differences may arise from the f(z) integration path, Omega_m0 handling, estimator normalization, or prediction-table provenance. Therefore the Appendix H Delta BIC values should be interpreted only as within-pipeline sensitivity diagnostics and should not be compared directly to the headline Delta BIC = -2.108.

mu0	chi2 exact	Delta BIC vs LCDM	mean E_G pred.
0.000000	27.044782	0.000000	0.422743
0.050000	24.933544	-2.111238	0.418340
0.100000	23.153051	-3.891731	0.414047
0.150000	21.679624	-5.365157	0.409859
0.172759	21.104666	-5.940116	0.407987
0.200000	20.491439	-6.553343	0.405774
0.250000	19.568358	-7.476424	0.401785
0.300000	18.891784	-8.152998	0.397891

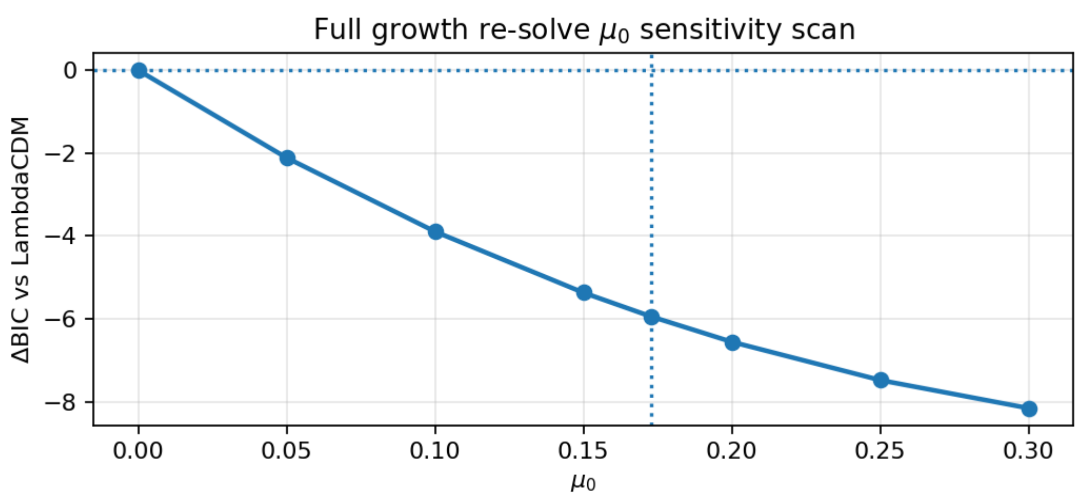

Figure H1. Full growth re-solve mu0 sensitivity scan. Negative Delta BIC values indicate that the neutral-lensing bridge prediction improves the diagonal pilot E_G comparison relative to LambdaCDM within this reconstructed verification pipeline. The vertical dotted line marks the frozen headline value mu0 = 0.172759. These values are not a replacement for the official C10 headline comparison.

Across the scanned range, every nonzero mu0 value remains negative in Delta BIC relative to LambdaCDM within this reconstructed pipeline, and the frozen headline value gives Delta BIC = -5.940116 in the re-solved diagnostic. The result supports the narrow claim that the pilot preference is not sharply fine-tuned to one carried-forward amplitude. However, the apparent improvement at larger mu0 should not be interpreted as a measured preference for larger amplitudes because the test is still diagonal-error only, not covariance-clean, and must be judged against external mu/Sigma constraints.